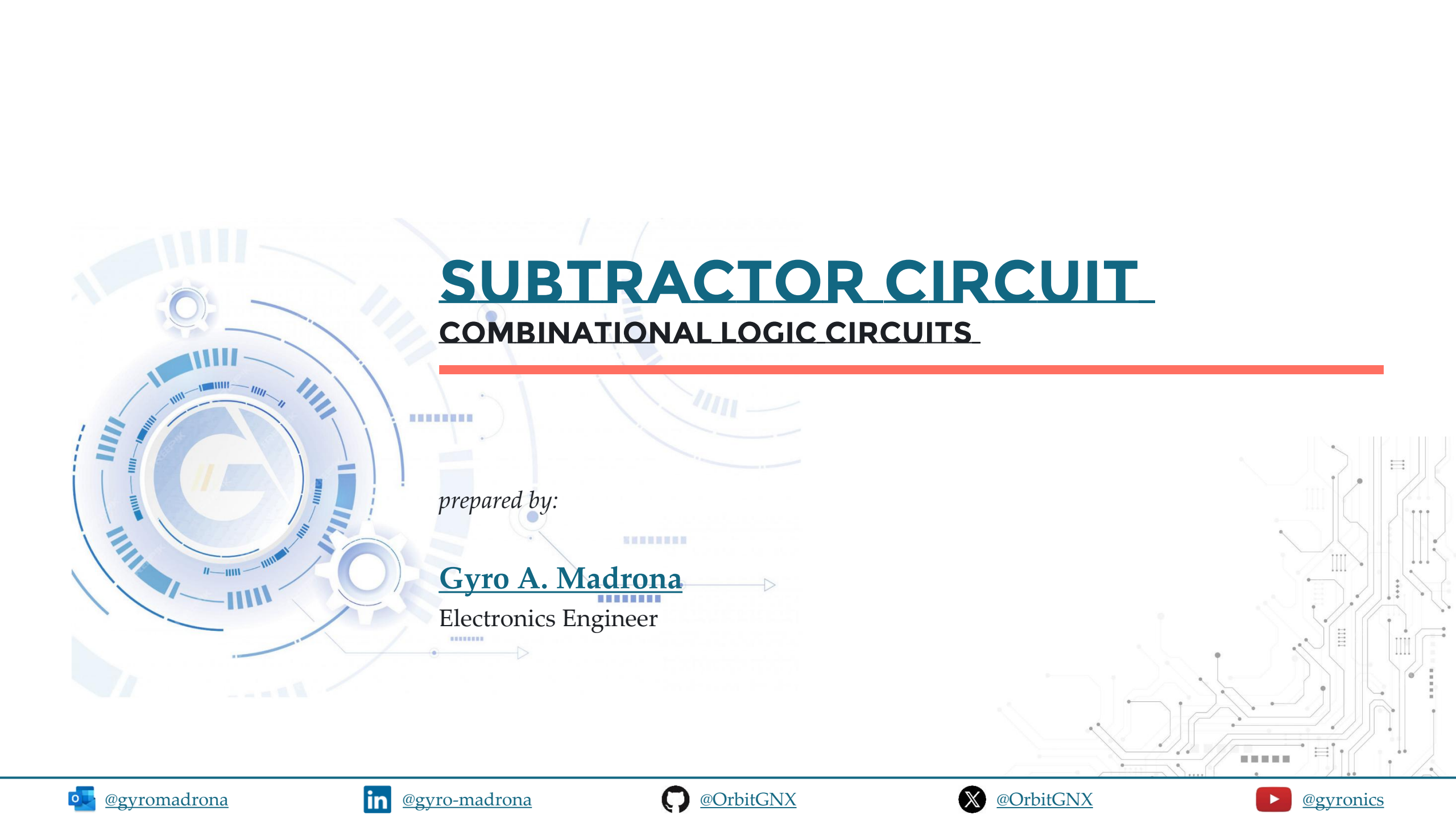




SUBTRACTOR CIRCUIT

COMBINATIONAL LOGIC CIRCUITS

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TOPIC OUTLINE

Signed-Magnitude Form

1's Complement Form

2's Complement Form



SUBTRACTOR CIRCUIT



SIGNED-MAGNITUDE

0101 (+5)

1101 (-5)

sign

0 - positive

1 - negative

Binary Addition

$$\begin{array}{r} + \quad 1101 \quad (-5) \\ \quad 0010 \quad (+2) \\ \hline \end{array}$$



$$\begin{array}{r} - \quad 101 \quad (5) \\ \quad 010 \quad (2) \\ \hline 1011 \quad (-3) \end{array}$$

$$\begin{array}{r} + \quad 0101 \quad (+5) \\ \quad 1010 \quad (-2) \\ \hline \end{array}$$



$$\begin{array}{r} - \quad 101 \quad (5) \\ \quad 010 \quad (2) \\ \hline 0011 \quad (+3) \end{array}$$

Drawback

To subtract the smaller number from the larger one, logic circuits for **comparison** and **subtraction** are needed.

1's COMPLEMENT

To obtain the 1's complement of a negative binary number, invert each bit – changing all 1s to 0s and all 0s to 1s.

Formula

$$K = (2^n - 1) - P$$

where:

K = negative number

P = positive number

Binary Addition

$$\begin{array}{r} 1010 \quad (-5) \\ + 0010 \quad (+2) \\ \hline 1100 \quad (-3) \\ 0011 \quad (1's) \end{array}$$

$$\begin{array}{r} 0101 \quad (+5) \\ + 1101 \quad (-2) \\ \hline 10010 \\ \text{adjustment} \rightarrow 1 \\ \hline 0011 \quad (+3) \end{array}$$

Drawback

In some cases, a correction is needed which amounts to an extra addition that must be performed.

1's COMPLEMENT

Equivalent Logic Circuit

To obtain the 1's complement of a negative binary number, invert each bit – changing all 1s to 0s and all 0s to 1s.

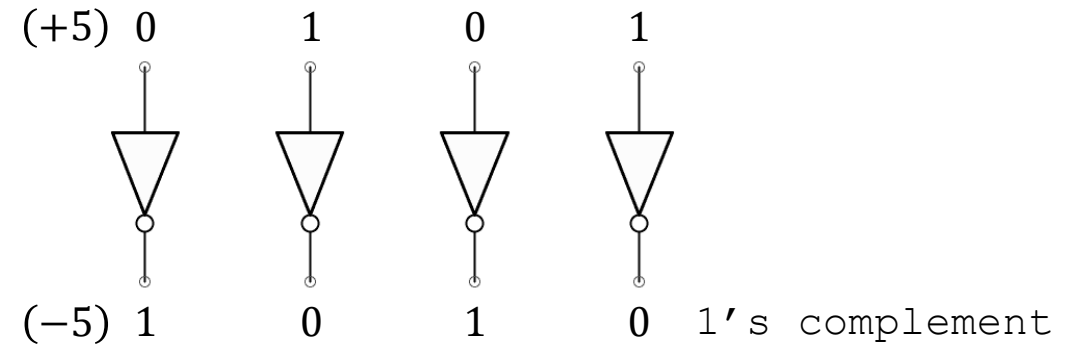
Formula

$$K = (2^n - 1) - P$$

where:

K = negative number

P = positive number



2's COMPLEMENT

To obtain the 2's complement of a negative number, first find its 1's complement (invert all bits), then add 1 to the result.

Formula

$$K = 2^n - P$$

where:

K = negative number

P = positive number

Binary Addition

$$\begin{array}{r} 1011 \quad (-5) \\ + 0010 \quad (+2) \\ \hline 1101 \quad (-3) \\ 0011 \quad (2's) \end{array}$$

$$\begin{array}{r} 0101 \quad (+5) \\ + 1110 \quad (-2) \\ \hline 10011 \quad (+3) \end{array}$$

ignore

Range

$$-2^{n-1} \text{ to } 2^{n-1} - 1$$

The addition process is the same, regardless of the signs of the operands.



2's COMPLEMENT

To obtain the 2's complement of a negative number, first find its 1's complement (invert all bits), then add 1 to the result.

Formula

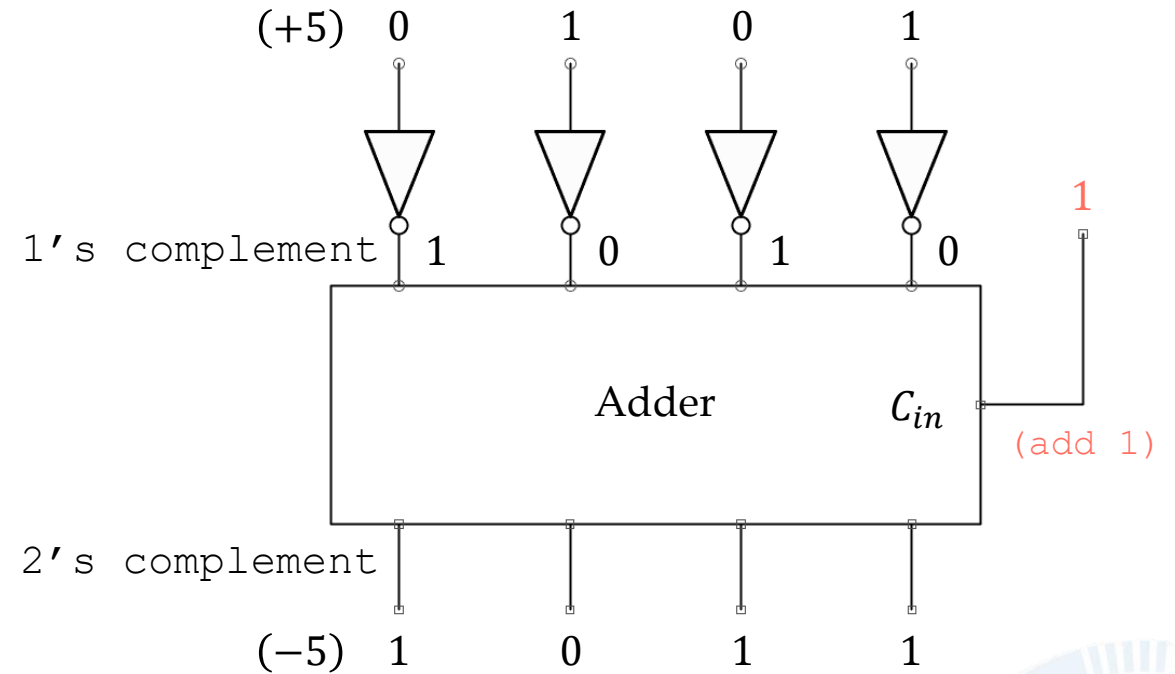
$$K = 2^n - P$$

where:

K = negative number

P = positive number

Equivalent Logic Circuit



EXERCISE

Create a block-level representation of a 2-bit binary subtractor using 2's complements method.

Solution



EXERCISE

Implement a 2-bit parallel binary subtractor using 2's complement method.

note

The use of XOR or XNOR gates is not allowed.

Solution



LABORATORY

